## EXP NO: 4B ENSEMBLE METHODS: RANDOM FOREST

### AIM

To implement a Random Forest classifier for a binary classification task using the Breast Cancer dataset, evaluate its performance using accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC, and identify the important features using feature importance.

### PROCEDURE

1. Import the required Python libraries.
2. Load the Breast Cancer dataset from Scikit-learn.
3. Separate the dataset into input features and target values.
4. Split the dataset into training and testing sets using stratification.
5. Create a Random Forest Classifier.
6. Train the Random Forest model using the training data.
7. Predict the target values for the test data.
8. Calculate accuracy, precision, recall and F1-score.
9. Generate the confusion matrix and classification report.
10. Calculate the ROC-AUC score and plot the ROC curve.
11. Determine the feature importance values from the trained Random Forest model.
12. Display the top 10 important features using a bar chart.
13. Perform hyperparameter tuning using GridSearchCV to obtain the best model parameters.

### CODE

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import f1_score, confusion_matrix, classification_report
from sklearn.metrics import roc_auc_score, roc_curve

data = load_breast_cancer()

X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")

X_train, X_test, y_train, y_test = train_test_split(
   X, y,
   test_size=0.20,
   random_state=42,
   stratify=y
)

rf = RandomForestClassifier(
   random_state=42,
   n_jobs=-1
)

param_grid = {
   "n_estimators": [100, 200],
   "max_depth": [None, 10],
   "min_samples_split": [2, 5],
```

```
    "min_samples_leaf": [1, 2]
}

grid = GridSearchCV(
    rf,
    param_grid,
    scoring="f1",
    cv=3,
    n_jobs=-1
)

grid.fit(X_train, y_train)

print("Best Parameters:")
print(grid.best_params_)

best_rf = grid.best_estimator_

y_pred = best_rf.predict(X_test)

y_prob = best_rf.predict_proba(X_test)[:, 1]

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred)
auc = roc_auc_score(y_test, y_prob)
```

```
cm = confusion_matrix(y_test, y_pred)

print("\nRandom Forest Test Results")
print("--------------------------")
print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
print("ROC-AUC  :", round(auc, 4))

print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

plt.figure(figsize=(6, 5))

sns.heatmap(
   cm,
   annot=True,
   fmt="d",
   cmap="Blues",
   xticklabels=data.target_names,
   yticklabels=data.target_names
)

plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
plt.title("Random Forest - Confusion Matrix")
plt.show()

importances = pd.Series(
   best_rf.feature_importances_,
   index=X.columns
)

top10 = importances.sort_values(
   ascending=False
).head(10)

plt.figure(figsize=(8, 6))

top10.sort_values().plot(kind="barh")

plt.xlabel("Importance")
plt.ylabel("Features")
plt.title("Top 10 Feature Importances")
plt.show()

fpr, tpr, thresholds = roc_curve(y_test, y_prob)

plt.figure(figsize=(7, 5))

plt.plot(
   fpr,
```

```
    tpr,
    label=f"Random Forest (AUC = {auc:.3f})"
)

plt.plot(
    [0, 1],
    [0, 1],
    linestyle="--"
)

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Random Forest")
plt.legend()
plt.show()
```

**OUTPUT:**

```
Best Parameters:
{'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 200}

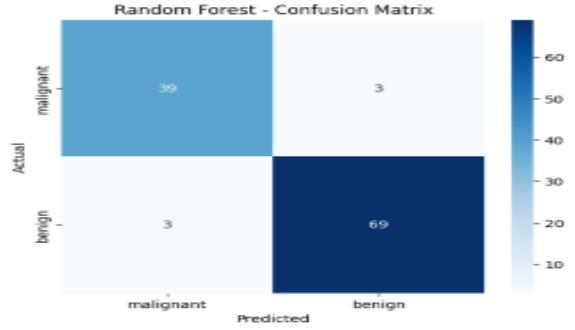
RANDOM FOREST - TEST RESULTS
----------------------------
Accuracy : 0.9474
Precision: 0.9583
Recall   : 0.9583
F1-Score : 0.9583
ROC-AUC  : 0.9934

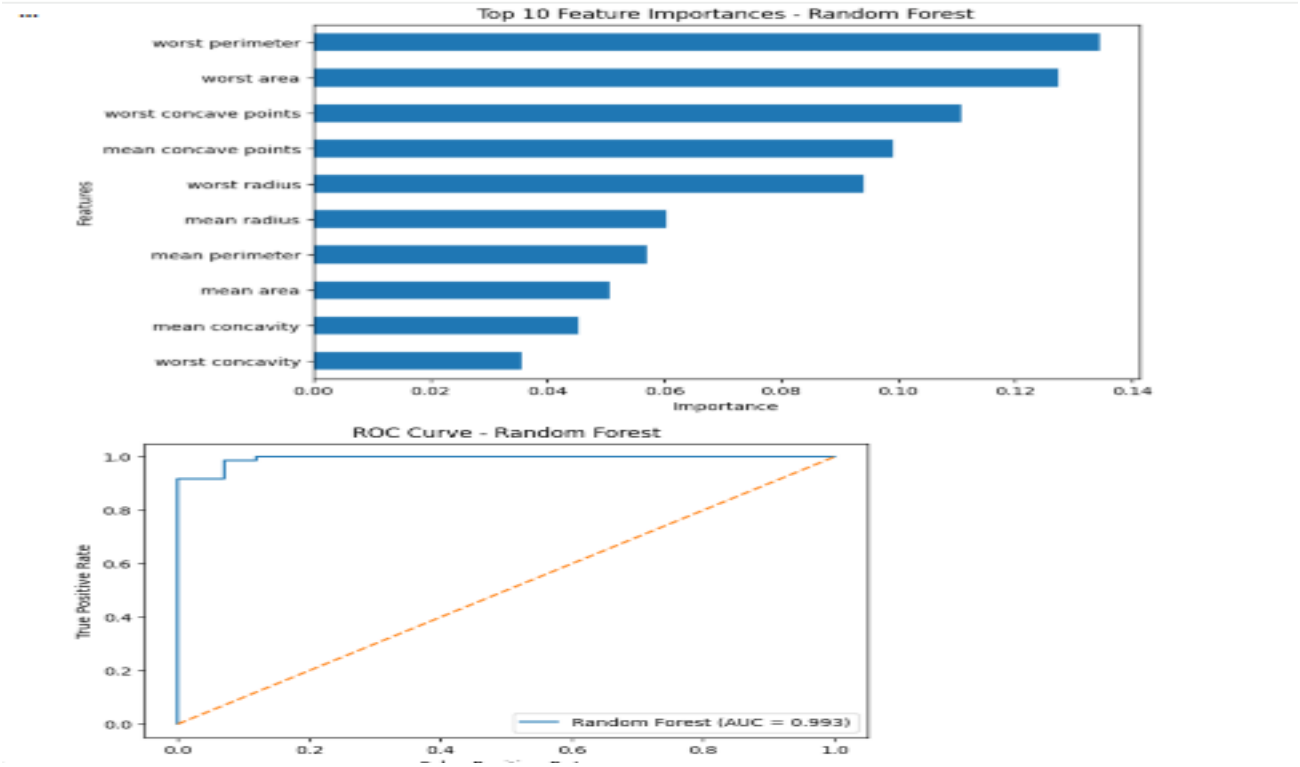
Confusion Matrix:
[[39  3]
 [ 3 69]]

Classification Report:
              precision    recall  f1-score   support

           0       0.93      0.93      0.93        42
           1       0.96      0.96      0.96        72

    accuracy                           0.95       114
   macro avg       0.94      0.94      0.94       114
weighted avg       0.95      0.95      0.95       114
```






**RESULT**

The Random Forest classifier was successfully implemented using the Breast Cancer dataset. The model was trained and tested successfully, and its performance was evaluated using accuracy, precision, recall, F1-score, confusion matrix and ROC-AUC. Feature importance was also analyzed to identify the most influential features. Hyperparameter tuning using GridSearchCV was successfully performed to obtain an improved Random Forest model.